Marketing Analytics

Text, Social Media, and Networks

By Nathan Yang

Session 07

A Customer-Service Decision

- A retailer wants to detect delivery complaints quickly.
- The document may be a review, sentence, or service conversation.
- The action is a review queue with limited staff capacity.

Text Preparation

- Keep negation and domain-specific terms when they affect meaning.
- Document language, duplicates, and boilerplate.
- Fit the vocabulary on training documents.

TF-IDF Representation

Weight a term by frequency in a document and rarity across documents.

$$\mathrm{tfidf}_{td} = \mathrm{tf}_{td} \log\left(\frac{N}{\mathrm{df}_t}\right)$$

Tokenization, smoothing, and normalization are implementation choices that need a record.

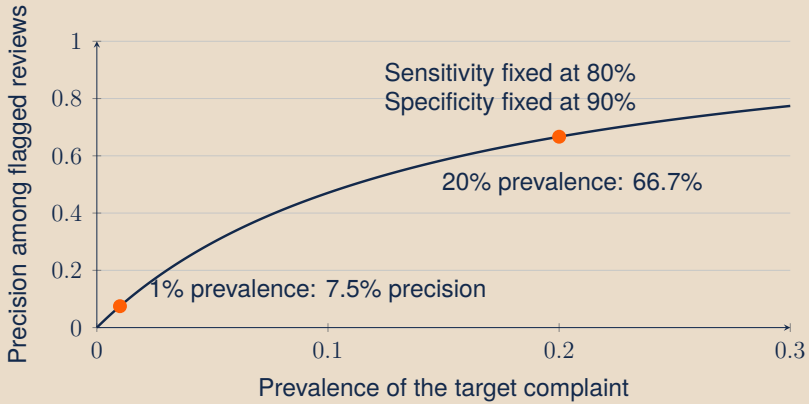
Embeddings and Similarity

- Embeddings place text in a numerical representation.
- Nearby descriptions may share meaning or usage.
- A useful similarity measure still requires domain validation.

Labels, Sentiment, and Topics

- Sentiment measures an evaluative construct.
- Aspect labels distinguish the object of a complaint.
- Topics summarize recurring term patterns under a model.

Rare Complaints Lower Precision



Synthetic classifier. Sensitivity is 80% and specificity is 90%.

Precision Depends on Prevalence

Let π be complaint prevalence, s sensitivity, and c specificity.

$$\Pr(Y = 1 \mid \widehat{Y} = 1) = \frac{s\pi}{s\pi + (1 - c)(1 - \pi)}$$

False positives can dominate a queue when the target condition is rare.

Expected Queue Composition

In 10,000 synthetic reviews, complaint prevalence is 1%.

```
n, prevalence = 10000, 0.01
sensitivity, specificity = 0.8, 0.9
tp = n * prevalence * sensitivity
fp = n * (1-prevalence) * (1-specificity)
precision = tp / (tp + fp)
assert round(precision, 4) == 0.0748
```

The expected queue contains 80 true complaints and 990 false positives.

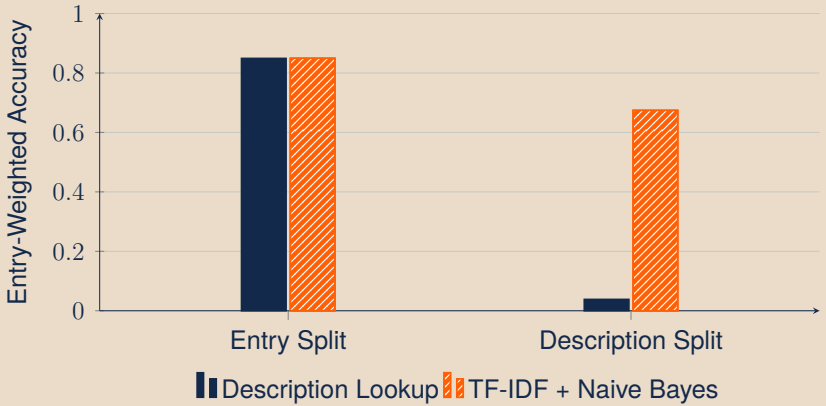
Human and LLM Annotation

- Write a label definition with difficult boundary examples.
- Measure agreement and adjudicate disagreements.
- Assess a held-out sample with human labels.
- Record the model, prompt, and response format.

mHealth Food Descriptions

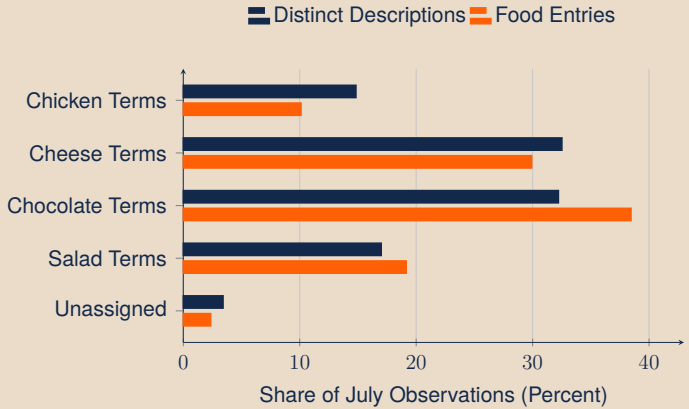
- Descriptions can recur across many users and dates.
- A repeated-description split evaluates familiar language.
- A held-out-description split tests generalization to new language.

Familiar and New Text Are Different Tests



mHealth text classification uses the stated training and evaluation splits.

Topic Shares Need a Denominator

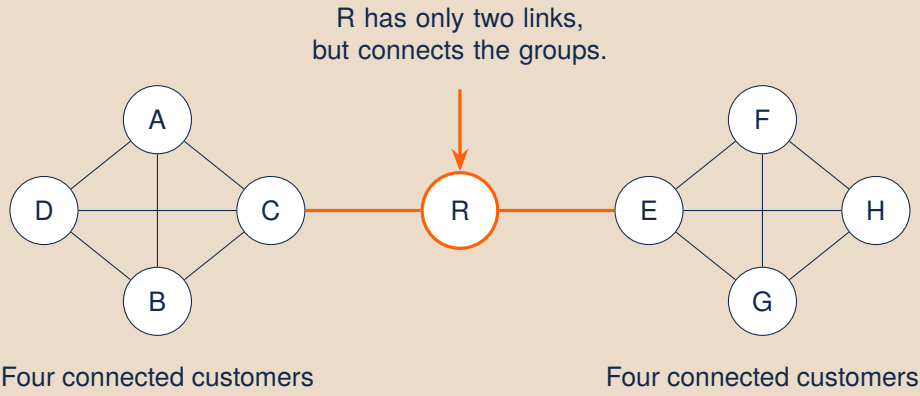


mHealth topic summaries compare documents, entries, and users under defined weights.

Network Objects

- Nodes identify customers or accounts.
- Edges represent a defined relationship in a time window.
- Degree counts links, while brokerage concerns paths between groups.

Few Links Can Connect Separate Groups



Synthetic undirected network. R has only two links.

Coding Exercise: A Label Assessment

- Build a transparent keyword or TF-IDF baseline.
- Compare precision and recall on familiar and new descriptions.
- Inspect false positives and disagreements by label.

Engagement and Commercial Value

- Likes and comments measure platform responses.
- Sales and contribution require separate outcome records.
- Network similarity can reflect common interests or common exposure.

Authenticity of Reviews and Endorsements

- A generated product description and a claimed customer review serve different roles.
- A fabricated experience misrepresents the source of the evidence.
- Keep sponsored relationships and substantive editing traceable.

Text-Analysis Discussion

- Which label disagreement would change the service decision?
- How does the expected queue change when prevalence falls?
- What comparison would distinguish exposure from influence?